

ICDFA MODULE 4 – LAB 04

INTRODUCTION TO SUPERVISORY CONTROL AND DATA

ACQUISITION (SCADA)

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Lab: Module 4 – Lab 04: Introduction to SCADA

System: OpenPLC + FUXA SCADA/HMI

Process: Tank Fill Simulation

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Introduction

Supervisory Control and Data Acquisition (SCADA) systems are used for process monitoring and control over industrial processes. The SCADA system architecture is mainly composed of field devices, a programmable logic controller (PLC), an industrial communication network, a SCADA application/Human-Machine Interface (HMI), and an operator.

The laboratory exercise aimed to experiment with a PLC and SCADA/HMI usage in process control through OpenPLC and FUXA software. A Tank Fill simulation was conducted to observe process control, PLC outputs, Modbus TCP communication, and supervisory visualization.

The laboratory exercise aimed to observe the process control, PLC outputs, Modbus TCP communication, and supervisory visualization. The experiment procedures mainly involved running the OpenPLC runtime, running the Tank Fill simulation, communicating with the FUXA SCADA system,

inspecting process tags under configuration, and identifying how process information is provided to an operator.

2. Lab Environment and Components

The laboratory environment consisted of the following components:

- **OpenPLC:** Programmable Logic Controller (PLC) runtime.
 - **FUXA:** SCADA/HMI application used for process visualization.
 - **Tank Fill Simulation:** Industrial process simulation controlled by the PLC.
 - **Modbus TCP:** Industrial communication protocol used between the PLC and SCADA environment.
 - **Tank Level (LIT):** Process measurement represented by the configured %IW80 / Modbus Input Register 80 tag.
 - **Fill Valve:** PLC output represented by %QX0.0 / Modbus Coil 0.
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3. Technical Execution

3.1 OpenPLC Runtime

The OpenPLC environment was started and the Tank Fill simulation program was loaded. The PLC runtime was successfully started and displayed:

Running: Tank_Fill_Sim-WORKING

The running status confirms that the PLC runtime was active and executing the configured Tank Fill control program.

Screenshot 1 – OpenPLC Runtime Running

SCREENSHOT

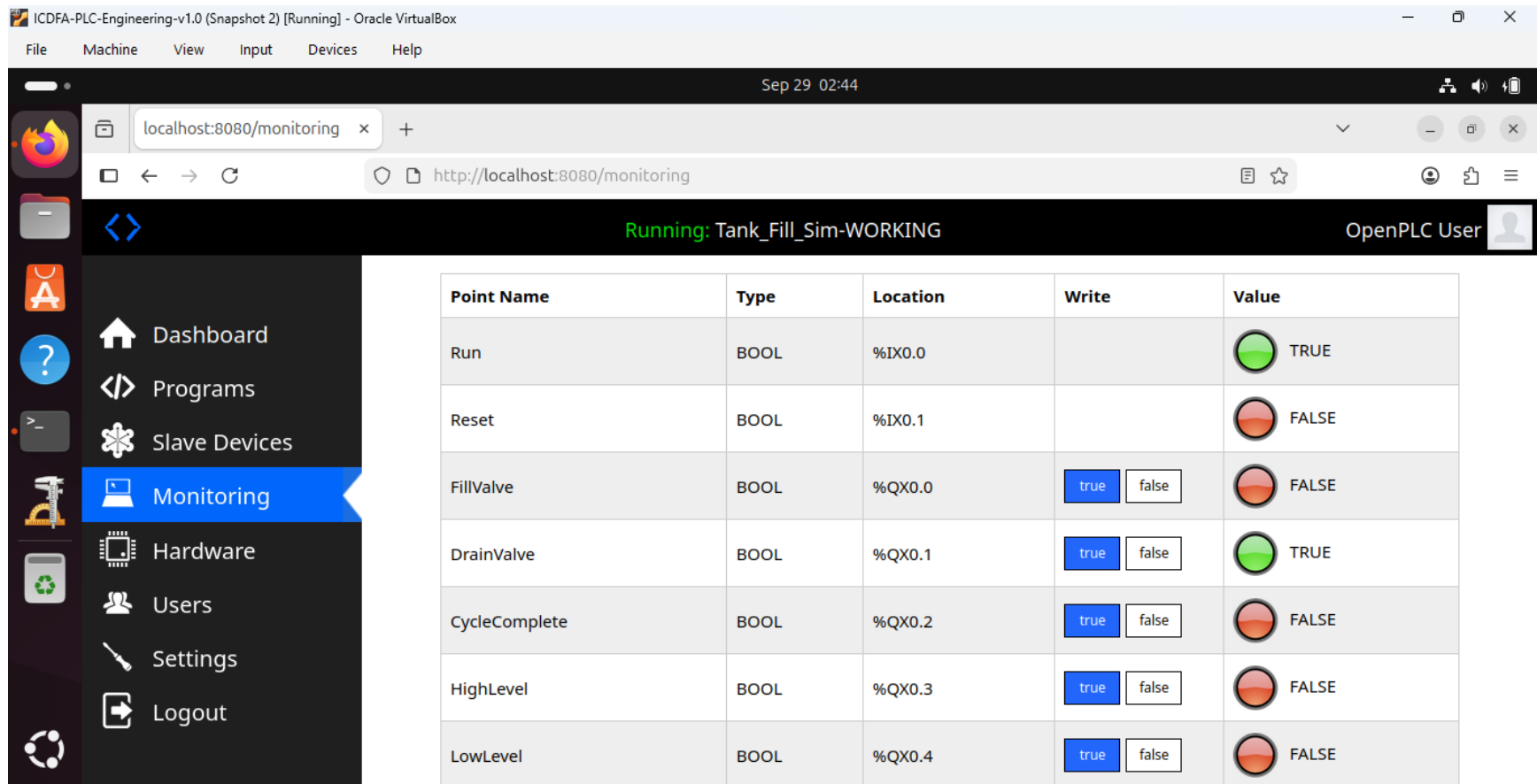
OpenPLC dashboard showing “Running: Tank_Fill_Sim-WORKING”.

The screenshot shows a web browser window titled "ICDFA-PLC-Engineering-v1.0 [Running] - Oracle VirtualBox". The browser's address bar shows "localhost:8080/dashboard". The page has a dark theme with a sidebar on the left containing icons for Dashboard, Programs, Slave Devices, Monitoring, Hardware, Users, Settings, and Logout. The main content area displays the status "Running: Tank_Fill_Sim-WORKING" in green text. Below this, the "Dashboard" section shows the following information:

- Status:** Running
- Program:** Tank_Fill_Sim-WORKING
- Description:**
- File:** 664729.st
- Runtime:** 1

The "Runtime Logs" section is visible below the dashboard information, showing a log entry:

```
<!DOCTYPE html>
<html>
```



3.2 FUXA SCADA/HMI

The FUXA SCADA/HMI environment was opened and the Tank Fill project was loaded. The connection indicator showed an active/green connection.

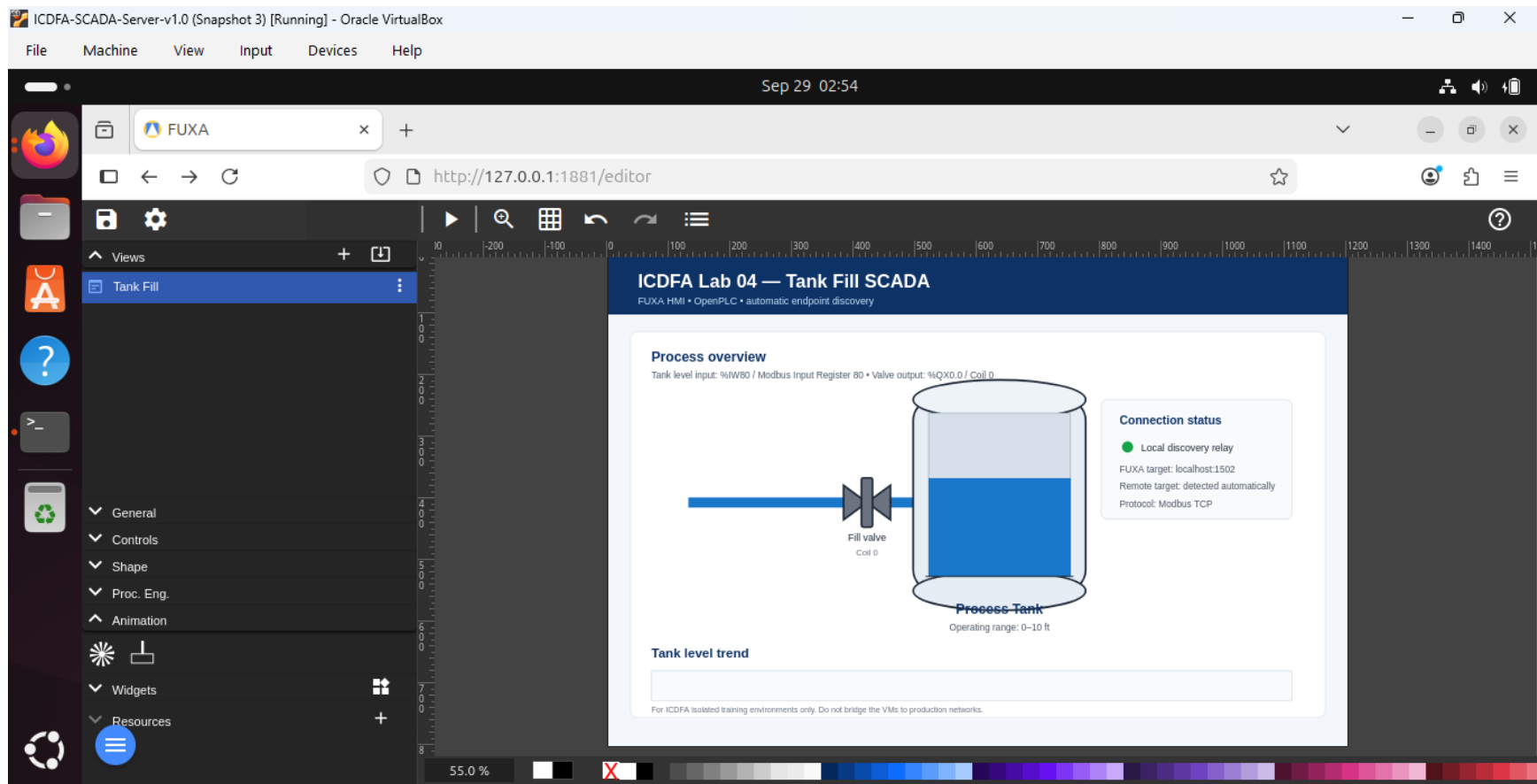
During operation, the FUXA interface displayed the PLC process states. The **Run** indication was TRUE and the **FillValve** indication was TRUE.

This demonstrated that the SCADA/HMI was receiving process/output information from the PLC environment.

Screenshot 2 – FUXA Tank Fill HMI

SCREENSHOT

Figure 2: FUXA Tank Fill HMI displaying the process interface and PLC status indications.



3.3 Process Overview and Tag Mapping

The FUXA Process Overview displayed the configured process variables used by the Tank Fill project.

The configured mappings included:

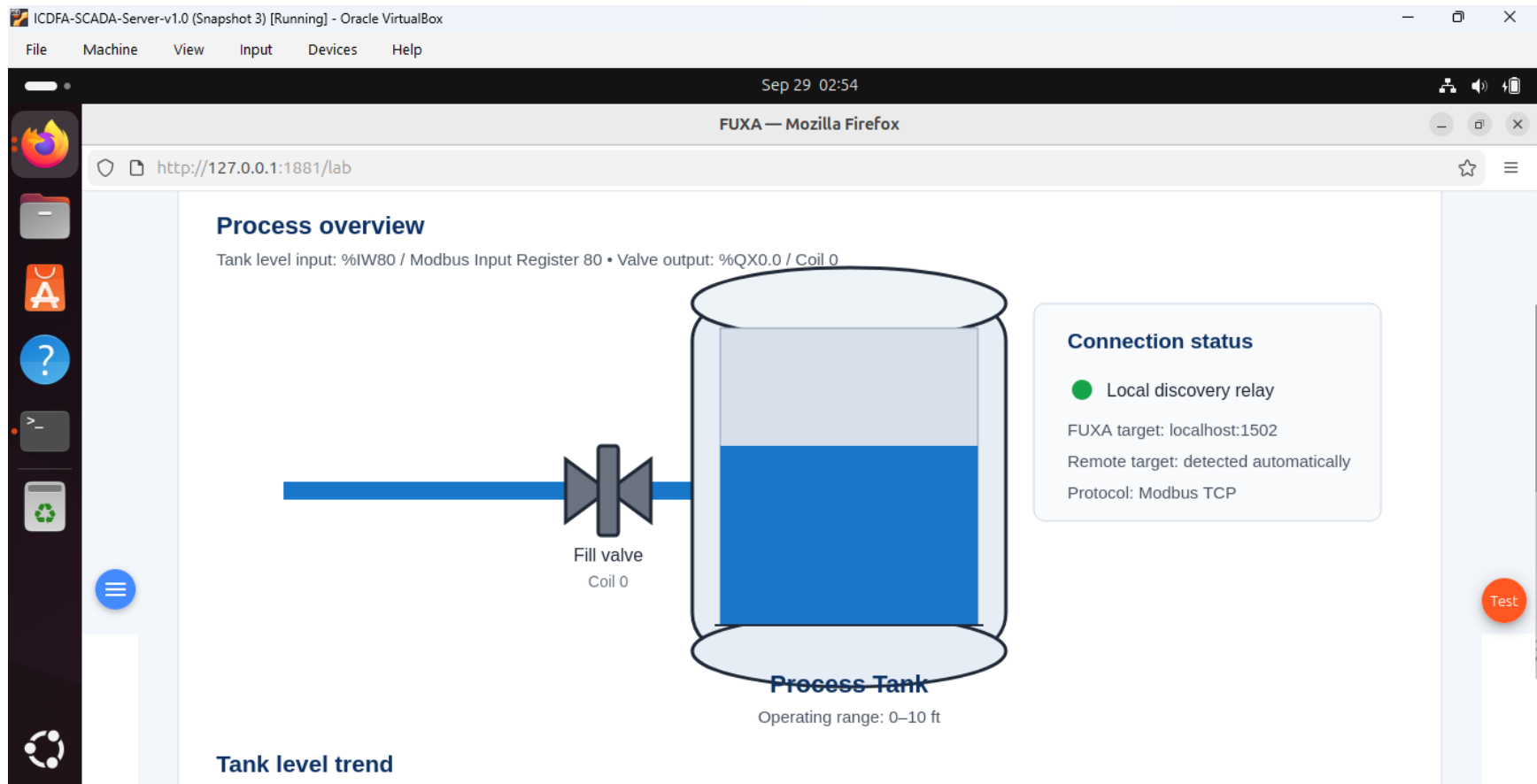
- **Tank Level (LIT):** %IW80 / Modbus Input Register 80
- **Valve Output:** %QX0.0 / Modbus Coil 0
- **Tank Operating Range:** 0–10 ft

These mappings demonstrate how the SCADA project associates process information with PLC/Modbus addresses.

Screenshot 3 – FUXA Process Overview

SCREENSHOT

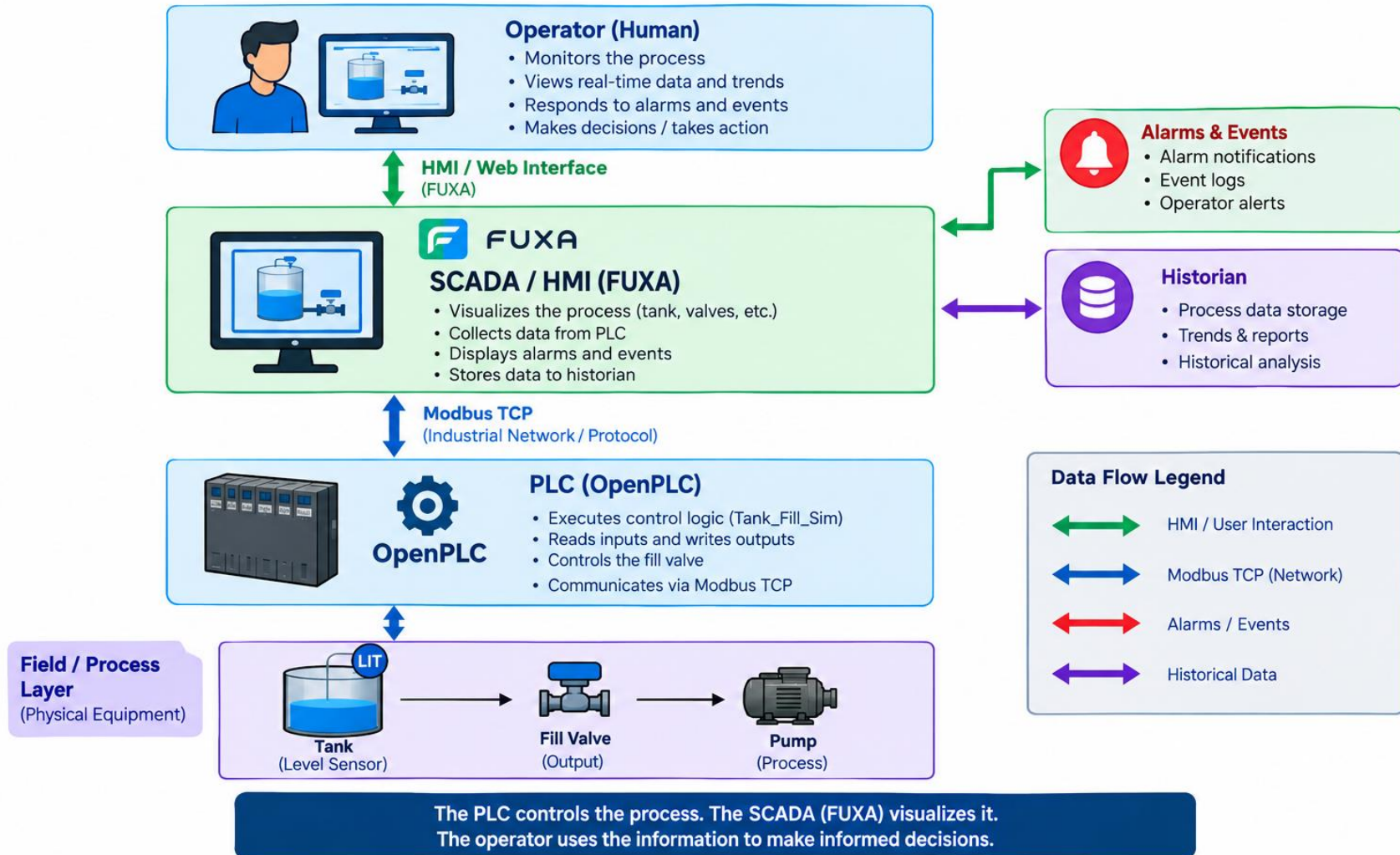
Figure 3: FUXA Process Overview showing Tank Level and Valve tag mappings.



4. SCADA Architecture

The architecture of the Tank Fill SCADA system can be represented as:

SCADA Architecture – Tank Fill Lab (OpenPLC + FUXA)



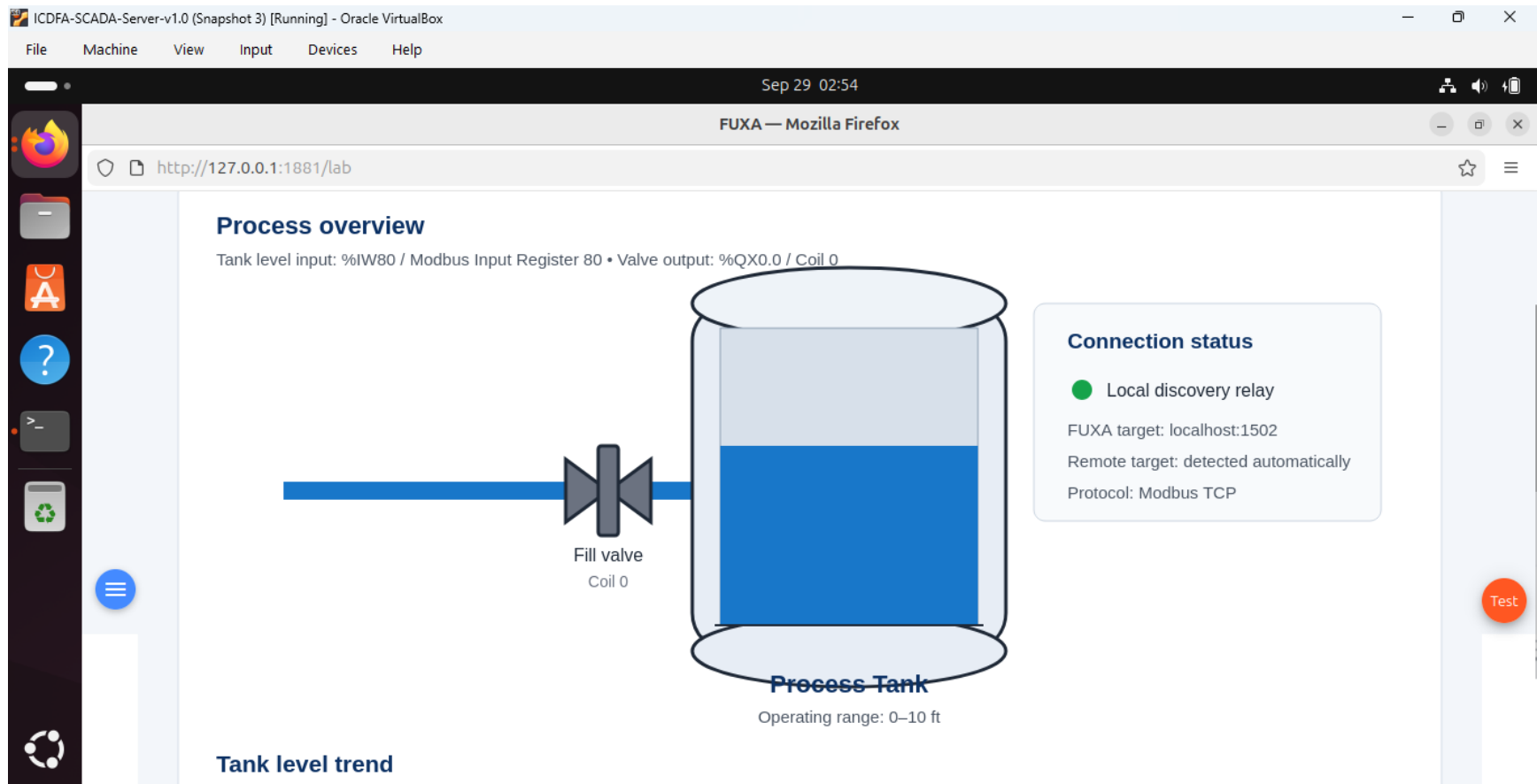
The field/process layer is the physical tank and process equipment. The PLC carries out the programmed control logic and controls process outputs such as the fill valve. Modbus TCP provides communication between the control and supervisory layers. FUXA provides visualization and supervisory monitoring for the operator.

Alarms and events are associated with the supervisory layer, while process data can be collected and historized for trend analysis and reporting.

Screenshot 4 – SCADA Architecture Diagram

THE SCADA ARCHITECTURE DIAGRAM

Figure 4: SCADA architecture showing the relationship between the field process, PLC, industrial network, SCADA/HMI, and operator.



5. PLC Control Logic and SCADA/HMI Visualization

The Programmable Logic Controller (PLC) is tasked with controlling the industrial process. The OpenPLC runtime is executing the Tank_Fill_Sim control program that contains the PLC code, which defines the process's input-output relations. The PLC is also responsible for controlling process equipment, such as the fill valve, and determining the tank's operating state. Thus, it can be concluded that the PLC is the primary actor in the industrial process, ensuring real-time and deterministic control over the system.

FUXA is the SCADA/HMI application, which is communicating with the PLC through the industrial network using the Modbus TCP protocol. The HMI component is providing process information to the operator in a human-readable form, which means that he can comprehend the current state of the process or tank, for example, the level percentage, and the status of the valves.

The main difference between the control and visualization software is that the former (PLC) is actually controlling the process, while the latter (SCADA/HMI) is only presenting information to the operators and allowing them to interact with the system in a human-friendly manner. The SCADA application is not able to substitute the control logic implemented in the PLC. The operators need the HMI because reading the values from the PLC's registers is not convenient, and they must have a higher-level view of the process. A visualization of the process makes operators aware of what is happening in the system, and critical information like limits, alarms, and events keeps them updated about the current state of the process.

6. Knowledge Check

6.1 What is the primary purpose of an HMI?

The primary purpose of an HMI (Human-Machine Interface) is to provide an interface through which an operator can monitor and interact with an industrial process. It presents process information such as tank level, valve status, alarms, and operating conditions in a clear and understandable format.

6.2 What is the difference between an alarm, an event record, and historized process data?

An **alarm** indicates an abnormal or potentially unsafe condition that requires operator attention. An **event record** documents something that happened in the system, such as a valve changing state or an operator action. **Historized process data** consists of recorded process measurements over time, such as tank-level values, which can be used for trends, analysis, and reporting.

6.3 Why does an operator need context rather than raw sensor values?

Raw sensor values do not always explain what is happening in the process. Context such as labels, operating ranges, status indicators, alarms, events, and trends helps the operator understand the meaning and significance of the data and make appropriate decisions.

6.4 What is the role of the PLC versus the SCADA application?

The **PLC** performs real-time control of the industrial process by executing programmed control logic and operating field devices. The **SCADA application** provides supervisory monitoring and visualization, collects and displays process information, and presents alarms, events, and historical information to operators. The PLC controls the process, while SCADA provides supervisory visibility and context.

7. Observations and Results

The OpenPLC runtime was successfully started and the Tank_Fill_Sim-WORKING program was shown as running.

The FUXA SCADA/HMI environment successfully displayed the Tank Fill process interface. The Run and FillValve indications were displayed as TRUE, demonstrating that process/output information was available through the SCADA interface.

The FUXA Process Overview also displayed the configured Tank Level and Valve mappings:

- Tank Level: %IW80 / Modbus Input Register 80
- Valve: %QX0.0 / Modbus Coil 0
- Engineering range: 0–10 ft

The Tank Level graphic itself did not visibly change during the observation period. Therefore, the evidence confirms the configured SCADA tags and the functioning PLC/output indication, but does not claim successful visual tank-level movement.

8. Analysis

The laboratory demonstrated the separation of responsibilities between industrial control and supervisory monitoring. The PLC is responsible for executing control logic and controlling process outputs, while the SCADA/HMI provides an operator-friendly representation of the process.

The use of Modbus TCP provides a communication mechanism through which process information can be exchanged between the control and supervisory environments. FUXA's Process Overview further demonstrates how SCADA variables are associated with PLC/Modbus addresses.

An important SCADA principle demonstrated by the exercise is that process data must be presented with sufficient context. An operator should not have to interpret raw register values without knowing what they represent, their engineering units, operating ranges, or whether a condition requires attention.

The exercise also demonstrates the importance of verifying both configuration and actual runtime behaviour. A configured tag does not necessarily prove that a process value is being updated correctly; operational testing and observation are required to validate the complete data path.

9. Conclusion

This laboratory provided practical experience with the basic architecture and operation of a SCADA system. The OpenPLC runtime successfully executed the Tank Fill simulation, while FUXA provided the SCADA/HMI interface for supervisory visualization.

The exercise demonstrated the relationship between the field/process layer, PLC control layer, industrial communication network, SCADA/HMI layer, and operator. It also reinforced the distinction between PLC control logic and SCADA visualization.

The configured FUXA process tags, including the Tank Level and Valve mappings, provided an example of how SCADA systems associate process information with PLC/Modbus addresses. Overall, the laboratory improved understanding of

PLC-based control, SCADA monitoring, HMI visualization, industrial communication, alarms/events, and the importance of providing operators with meaningful process context.

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